Warning.--This document contains information affecting the national defense of the United States within the meaning of the Espionage Laws, Title 18, U.S.C., Sections 793 and 794. The transmission or the revelation of its contents in any manner to an unauthorized person is prohibited by law.

MIL-R-11043A (ORD)
23 October 1951
SUPERSEDING
MIL-R-11043 (ORD)
28 March 1951

MILITARY SPECIFICATION

# ROCKET, SMOKE, WP, 3.5 INCH, T127E2, HEAD ASSEMBLY: METAL PARTS FOR

## 1. SCOPE

1.1 This specification covers the head metal parts assembly for Rocket, Smoke, WP, 3.5 Inch, T127E2.

## 2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 The following drawing and all drawings and specifications referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

ORDNANCE CORPS DRAWING
P-80855 - Rocket, Smoke, WP, 3.5 Inch, T127E2. Head Metal Parts Assembly And Details.

(Copies of specifications and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Material.--Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 The metal parts shall comply with all requirements specified on applicable drawing and in applicable specifications.

3.3 Workmanship.

3.3.1 All parts shall be free of chips, dirt, grease, and other foreign terial. The cleaning method used shall not be injurious to any of the parts shall the parts be contaminated by the cleaning agents used.

3.3.2 The interior and exterior surfaces shall be free of burrs, rust, scars, and scratches.

3.3.3 The interior and exterior coatings specified on the drawings shall ied in such a manner as to insure an even distribution and a thorough

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:02Z
Check the source to verify that this is the current version before use.

FSC 1340 A

3.3.4 Flux.--All flux remaining in the body cavity after brazing shall be removed by an accptable cleaning method.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.--A lot shall consist of 10,000 metal parts assemblies, maximum. A smaller quantity may constitute a lot at the discretion of the inspector. One thousand, plus or minus 250, shall constitute the first lot manufactured on any contract except when a new contract immediately follows the manufacture of satisfactory material of the same character. Each lot of head metal parts assemblies shall contain components made by one manufacturer in accordance with the same draving and draving revision, and same specification and specification revision

4.2 Inspection.--Inspection shall be as specified in Specification MIL-G-2550, and Specification QQ-M-151.

4.3 Internal air pressure test.

4.3.1 Procedure.--Each head metal parts assembly in the lot shall be subjected to the internal air pressure test specified on the draving. The pressure shall be applied at the threaded end, after the head is assembled in a fixture similar to that shown on figure 1 (see 6.1).

4.3.2 Acceptance.--Each head metal parts assembly shall be considered acceptable provided it withstands, without a drop in pressure, the internal air pressure test specified on the draving. If any assembly fails to comply with this requirement, it shall be rejected subject to correction and resubmission for test.

4.4 External air pressure test.

4.4.1 Procedure.--Each head metal parts assembly shall be subjected to the external air pressure test specified on the draving. The pressure shall be applied to the head after assembly in a fixture similar to that shown on figure 1 (see 6.2).

4.4.2 Acceptance.--Each head metal parts assembly shall be considered acceptable provided it withstands, without a drop in pressure, the external air pressure test specified on the draving. If any assembly fails to comply with this requirement, it shall be rejected subject to correction and resubmission for test.

## 5. PREPARATION FOR DELIVERY

5.1 Packing, labeling and marking.--Packing, labeling and marking shall be in accordance with Specification MIL-P-10025.

5.2 Data cards.--Data cards, 5 by 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of accepted metal parts to provide the information specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Office, Chief of Ordnance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:02Z
Check the source to verify that this is the current version before use.

DATA CARD

Kind_____ Lot no._____
Quantity in lot_____
Dwg no.*_____ Spec no.*_____
Mfd by_____ Date_____
Contract_____ Order no._____
Packed_____ in a box
Remarks_____

COMPONENTS

| | Model | Mfr | Lot No. | Year Mfd | Dwg No. * |
|---|---|---|---|---|---|
| Body and Union Assembly | | | | | |
| Ogive | | | | | |

Certified by_____, Inspector.
(Print here name of inspector, district, arsenal or loading plant, and sub-district where applicable.)

* Include revision and amendments.

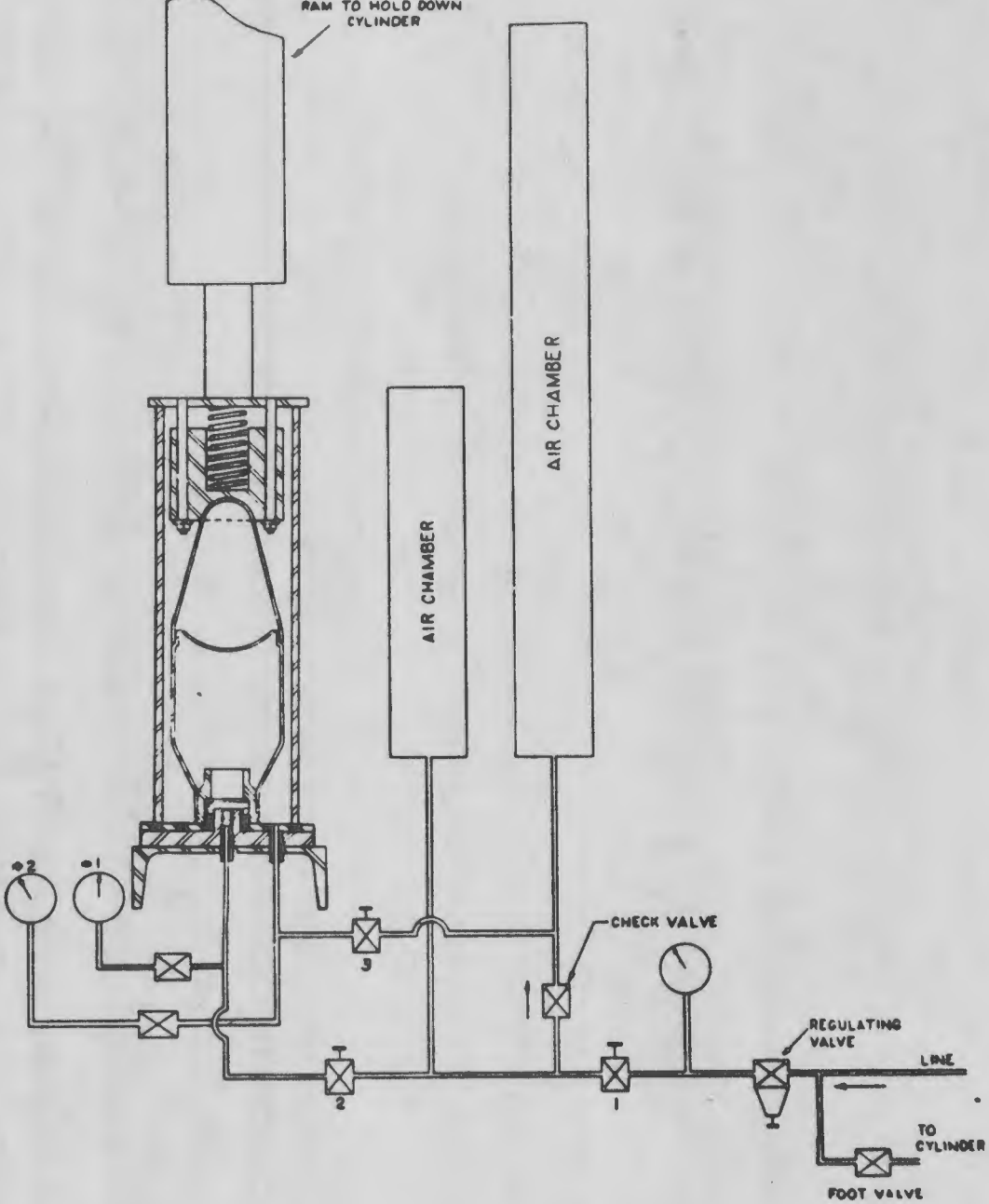
6. NOTES

6.1 The sensitivity of the test fixture shall be such that a test head assembly, containing a .0135 diameter hole drilled in the center of the dome section of the body, when tested in accordance with 4.3.1, will produce a pressure drop of 2 psi per second minimum for the first 5 seconds after the 90 psi has been introduced into the head.

6.2 The sensitivity of the test fixture shall be such that a test head, containing a .0135 diameter hole drilled in the ogive 2 inches from the apex, when tested in accordance with 4.4.1, will produce a pressure drop of 2 psi per second minimum for the first 5 seconds after the 90 psi has been introduced into the head.

Custodian:
Army Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:02Z
Check the source to verify that this is the current version before use.




STEPS:-

1 CLAMP ROCKET HEAD IN CASING USING FOOT VALVE.
2 OPEN VALVE NUMBER 1 ABOUT 5 SEC
3. OPEN VALVE NUMBER 2 ABOUT 5 SEC. HOLD FOR 60 SECONDS AFTER VALVE NUMBER 2 IS CLOSED.
(IF THERE IS AN INTERNAL LEAK FROM BODY TO OGIVE OR FROM BODY TO ATMOSPHERE, THE NUMBER 1 GAGE WILL READ BELOW NORMAL; THIS REJECTS HEAD WITHOUT FURTHER TEST.) IF TEST IS SATISFACTORY, CONTINUE.
4 OPEN VALVE NUMBER 3 ABOUT 5 SEC, HOLD FOR 60 SEC AFTER VALVE NUMBER 3 IS CLOSED. IF #2 GAGE DROPS BELOW NORMAL, THERE IS A LEAK BETWEEN OGIVE AND ATMOSPHERE
5 RELEASE ROCKET WITH FOOT VALVE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T15:02Z
Check the source to verify that this is the current version before use.

SUGGESTED APPARATUS FOR INTERNAL AND EXTERNAL AIR PRESSURE TESTS OF 3.5 ROCKET.

TIZER HEAD ASSEMBLY